

UNDER RUBBL

SURVIVAL OS

Tech that works when everything else fails.

v2.4.1

People are trapped. Their phone is their only lifeline.



NO SIGNAL

Cellular networks collapse in disasters. Standard apps become useless.



PANIC KILLS

Hyperventilation and shock consume oxygen 3x faster. No guidance = worse outcomes.



CAN'T BE FOUND

Rescuers can't locate survivors who can't signal. 72hr window is critical.

One app. Zero infrastructure needed.

UnderRubbl is a Progressive Web App that runs offline, needs no install, and keeps survivors alive until rescue.



SOS ACOUSTIC BEACON

4000Hz sine-wave Morse S-O-S pattern designed to penetrate debris. Real Web Audio API — no network needed.



AI CRISIS PSYCHOLOGIST

Dr. Aria — Claude Sonnet AI trained on trauma psychology. Calm conversation, box breathing, panic protocols.



LIVE HAZARD MONITOR

Live USGS earthquake GeoJSON + weather threat matrix. Know what's happening above ground.



BATTERY MANAGEMENT

Browser Battery API with conservation protocols at 50%→30%→15%→5% thresholds.



MEDICAL ID

Blood type, allergies, medications — instantly readable by first responders. Persistent across reloads.



SURVIVAL TIMER

Tracks exact time since entrapment. Classifies phase: Acute → Serious → Severe → Critical.

DR. ARIA

AI Crisis Psychologist

- ONLINE | DR. ARIA — Trauma-Informed AI Psychologist

Maria, I'm here with you. Take one slow breath. You are not alone — rescuers are searching right now. What's your most immediate need?

I can't breathe properly. I'm panicking.

That's okay — panic is your body protecting you. Let's slow it down together. Inhale through your nose for 4 counts... hold... exhale for 4. Repeat with me.



Trauma-Informed System Prompt

Claude Sonnet trained with crisis psychology protocols. Validates emotions before instructions.



Box Breathing Widget

Animated 4-4-4-4 cycle with visual inhale/exhale circle. Activates parasympathetic nervous system.



5-4-3-2-1 Grounding

Sensory grounding protocol built into quick-action buttons. One tap for immediate help.



Offline Fallback

When API unavailable, static crisis protocols still guide the user through core techniques.

REAL-TIME HAZARD MONITORING



SEISMIC ACTIVITY – USGS LIVE FEED

MAJOR**M6.2**

47km NE of local epicenter
2m ago

MODERATE**M4.8**

12km from your location
18m ago

MINOR**M3.1**

31km SE of city center
1hr ago



USGS GeoJSON API

Pulls real M4.5+ earthquake data from the US Geological Survey feed. Graceful fallback to simulated data when offline.



WEATHER THREAT MATRIX

Structural fire risk, particulate hazard, rain forecasts. Each alert has color-coded severity.



AFTERSHOCK PROTOCOL

Built-in action card: what to do during and after shaking. Covers gas, airway, movement decisions.



ONE-TAP REFRESH

Users can manually refresh the feed. Timestamp shows data freshness for trust.

Built lean. Built to survive.

*No heavy frameworks. No dependencies that can fail.
Every byte justified for emergency use.*

FRONTEND	HTML5 + Tailwind CSS	Zero-install, works in any browser
AUDIO	Web Audio API	Hardware-level 4000Hz SOS beacon
HARDWARE	Battery API + Wake Lock	Real power monitoring, screen stays on
AI	Claude Sonnet (Anthropic)	Trauma-informed crisis conversation
DATA	USGS GeoJSON API	Live earthquake feed, no key needed
OFFLINE	Service Worker PWA	All core features work without internet
STORAGE	localStorage	Medical ID persists across restarts

How we used AI.

As required by the hackathon rules, here is a full breakdown of AI tools used in this project.

Claude Sonnet

IN-APP AI

Powers Dr. Aria, the crisis psychologist. Custom trauma-informed system prompt authored by the team.

Claude claude.ai

CODE GENERATION

Generated the majority of the HTML/CSS/JavaScript codebase. All logic reviewed and understood by the team.

Claude claude.ai

PRESENTATION

Generated this presentation script. Content and structure directed by the team.

Human Team

DIRECTION & QA

All design decisions, feature selection, prompt engineering, and hackathon alignment done by the team.

THE TEAM "DIGITAL CROSSROADS"



MYKYTA LOBKO

Co-Founder & AI Engineer
Claude AI integration
Prompt engineering
Frontend development



OLEKSANDR ZAIETS

Co-Founder & Lead Developer
Hardware architecture
Sensor integration
Embedded software

EVERY SECOND COUNTS.

*UnderRubbl doesn't wait for infrastructure.
It works now. In the dark. Under the rubble.*

4000

Hz BEACON FREQ

0

MB INSTALL SIZE

72hr

CRITICAL WINDOW



PROJECT



GITHUB